

LAB 3 — ARCHITECTURE SELECTION AND JUSTIFICATION

Self-Service Coffee Kiosk System

Architecture Selection

We chose Layered Architecture for the Self-Service Coffee Kiosk System.

Architectural Choice

The system is divided into three layers: Presentation, Business and Data. The Presentation Layer contains the User Interface Component for handling touch-screen interactions. The Business Layer contains the Order Manager, Payment Service and Receipt Printer components for handling ordering, payment and receipt operations. The Data Layer contains the Database Component for storing menu and pricing information.

Reason 1

Layered Architecture provides clear separation of concerns. The user interface, order processing, payment processing and data storage are separated into different components, making the system easier to understand, maintain and modify.

Reason 2

The coffee kiosk has simple and well-defined functionality such as selecting a coffee, selecting a size, making a credit-card payment and printing a receipt. Therefore, Layered Architecture is suitable without introducing the operational complexity of a Microservices Architecture.

Security Advantage

Payment processing is isolated in the Payment Service Component. The User Interface does not directly access the database, which helps restrict access to system data and keeps responsibilities separated.

Performance Benefit

The system can retrieve menu and pricing information through the business layer while keeping the presentation layer focused on touch-screen interaction. Separating responsibilities reduces unnecessary processing in the user interface.